Journal

Label	Type of work
R	Recherche
D	Documentation
P	Programming
O	Organization

Date	Work	Type	Time (min)
2025	-	-	-
10/20	Creating a document explaining the basic idea and inspiration behind the senior thesis. Consulted with biology teachers.	D	120
11/29	Filling out and submitting/uploading the preliminary contract.	O	10
12/16	First meeting with Mr. Leuthardt. Formulating a possible goal for the project.	O	30
12/30	Creating a GitHub repository for the code.	P	10
2026	-	-	-
01/14	Second meeting with Mr. Leuthardt. Discussing and completing the work agreement. Setting a goal to be achieved by the next meeting.	O	30
01/23	Submitting/Uploading the Work Agreement.	O	10
01/28	Configuring the repository for C++ (CMakeLists.txt, src/main.cpp). Importing some header files from another project. Installing Vulkan and GLFW.	P	120

01/28	Accidentally deleted everything permanently. Research on “inheritance” and “unions” in C++.	R	60
01/29	Reconfiguring the project. Creating the first elements: struct Vec<N, T> (specifically: Vec<2, T>, Vec<3, T>, Vec<4, T>), Fundamental types (i8_t, u8_t, i16_t, u16_t, i32_t, u32_t, i64_t, u64_t, f32_t, f64_t, cstr_t) Help: ChatGPT, Reddit, Stack Overflow	P, R	180
01/30	Continue working on the code and creating a new element: class Window (create and manage a GLFWwindow) Help: ChatGPT, Reddit, Stack Overflow, Vulkan Tutorial (Alexander Overvoorde)	P, R	120
01/31	Creating and Updating the Journal	D	60
01/31	Continue working on the code and creating new elements: class VulkanHandle (create and manage Vulkan-specific resources), struct Version (specify a version: x.y.z, compatible with Vulkan) Help: ChatGPT, Stack Overflow, Vulkan Tutorial	P, R	270
02/01	Researching on how to document C++ code and the documentation for window.hpp Help: ChatGPT , Stack Overflow	P, D, R	390
02/02	Documenting the Code	P, D	240

02/03	Continue working on the code and adding new features: Initializing and creating: VkPhysicalDevice, VkQueueFamilies, and VkDevice. Help: Vulkan Tutorial	P	180
02/04	Continue working on the code. Attempting to initialize the Vulkan graphics and presentation queues. Adding a new element: VkSurface Help: Vulkan Tutorial	P	120
02/05	Continue working on the code. Successfully initialized the VkSurface and created the graphics and presentation queues. Help: Vulkan Tutorial	P	240
03/03	Continue working on the code. Begin implementing VkSwapchainKHR. Adding the necessary elements. Help: ChatGPT, Stack Overflow, Vulkan Tutorial	P, R	120
03/04	Continue working on the code. Successfully initialized VkSwapchainKHR and began implementing VkPipeline. Added the necessary elements. Help: Vulkan Tutorial	P, R	180
03/05	Continued working on the code. Successfully initialized the graphics pipeline and implemented the VkCommandBuffer functionality and related elements. Successfully ran the program for the first time and rendered a single triangle. Optimized using the “frames-in-flight” functionality.	P, R	500

	Help: ChatGPT, Vulkan Tutorial		
03/07	Continue working on the code. Start implementing a vertex buffer. Help: Vulkan Tutorial	P, R	180
03/08	Continue working on the code. Switching from GLFW to SDL3. Adding a logging and debugging system for resource creation and errors. Help: ChatGPT, Stack Overflow, Vulkan Tutorial	P, R	240
03/09	Continue working on the code. Successfully implemented the vertex buffer and worked on various shaders. Help: Vulkan Tutorial	P, R	300
03/10	Continue working on the code. Adding a Vulkan push constant for the screen resolution (required by the shaders). Successfully implement a working circle shader. Help: Vulkan Tutorial	P, R	180
03/11	Continue working on the code. Experimenting with different fragment shaders and implement color blending. Help: ChatGPT, Vulkan Tutorial	P, R	120
03/12	Continue working on the code. Adding a function (DrawCircle()) to the Window class to draw a circle. Still has a few issues. Help: ChatGPT, Vulkan tutorial	P	120
04/03	Continue working on the code. DrawCircle() is working and running some initial tests. The circle renderer is mostly finished. Help: ChatGPT, Vulkan Tutorial	P, R, D	180

04/04	Continue working on the code. Adding a function (DrawRectangle()) to the Window class to draw a rectangle. Adding functions to draw circles and rectangles with various shader effects. Help: ChatGPT, Stack Overflow, Vulkan Tutorial	P	180
05/02	Continued working on the code. Deleting the existing chemical simulation elements and started working on biological cell components. Added a “Particle” and a “Compound” class and began implementing phospholipid functionality.	P, R	180
05/03	Initial success with phospholipids. The hydrophobic ends cluster together, forming small groups. Due to problems with the rotation calculation, the particles do not remain parallel to one another → no longer chains are possible yet. Adding a few adjustable constants to change the behavior of the particles and experiment. Help: ChatGPT	P, R	420
05/05	Modifying the interaction between hydrophilic and hydrophobic particles. Old: Linear scaling of attraction/repulsion based on distance. Collisions or overlaps must also be detected and prevented. New: Attraction/repulsion based on the “Lennard-Jones potential.” Prevents collisions through exponential repulsion at close range. The new approach solves the problem with rotation calculations, and for the first time, longer chains (more than 10 phospholipids) are formed.	P, R	240

	Help: ChatGPT, Wikipedia on the Lennard-Jones potential		
05/06 – 05/16	Adjusting the constants to allow a single chain to form without assistance. Improvements have been made. Adjusted values: maximum interaction distance between particles, hydrophobic attraction, hydrophilic attraction, repulsion between hydrophobic and hydrophilic particles, constants for linear and angular velocity, velocity damping over time. Furthermore, limiting velocity decay to minimum velocities to prevent all particles from coming to a stop. The total velocity of all particles still decreases slowly → accelerate all particles as soon as the total velocity falls below a certain value Help: ChatGPT	P, R	480
05/17	Catching up on the Journal from the past few days. Used a script to count the total number of lines in all source files: approximately 7,200. Help: ChatGPT	D	30
06/14	Considerations regarding the implementation of proteins, DNA/RNA, and their interactions. Further experiments on the phospholipid bilayer with different values.	P	45
06/15	Implementing a system to make it easier to navigate the simulation area. Zoom by scrolling with the mouse and move using the arrow keys.	P	120

06/17	Last meeting with Mr. Leuthardt before summer break.	O	15
06/21	Continue working on the implementation of proteins, RNA, and DNA. Help: ChatGPT, Stack Overflow	P	60
06/22	Cleaning up and organizing the program. Moving all elements in main.cpp (except the main() function) into header files. Global variables and constants → parameters.hpp, Particle class → particle.hpp, Compound class and associated elements → compound.hpp, remaining functions → functions.hpp. Adding a header file for all possible connections in the simulation (compound_structures.hpp). Counting the total number of lines in all source files again: approximately 7,500.	P	60
06/23	Synchronized the code and log with GitHub. Added a timer class (singleton) to manage time-related data and calculate the time elapsed in the simulation since a specific frame (important for subsequent protein dynamics). Added a third-party library to save the simulation data in JSON format. Help: ChatGPT	P	300

06/24	Adding a system that stores all important data from a running simulation and can reload stored data. Creating a hash map that stores all possible connections and can be used to create a new element based solely on its name (e.g. “phospholipid” and “protein_1”). First prototypes of functions that enable specific interactions between compounds (transcription, translation). First success in simulating protein biosynthesis. Help: ChatGPT, Stack Overflow	P	480
06/25	Improving the interaction system and adding a DNA replication feature. Adding another third-party library to parallelize the program using a thread pool. Further improvements and bug fixes in the code. Creating a special element that deletes everything it touches to better control the simulation. Implementing parallelization and improving the program’s speed. At approximately 500 connections: approx. 38 FPS (without parallelization) → approx. 75 FPS (with parallelization)	P	600
06/26	Improved interactions for transcription, translation, and DNA replication. Proteins now move along the DNA/RNA strand and form new bonds as soon as they reach the end. Added an optional lifetime for bonds.	P	360

	Restructuring the project into modules that are compiled separately → after a change, the entire code does not need to be recompiled. Reversing parallelization, since global variables cannot simply be modified by multiple threads. Code for parallelization has been commented out. Fixing various bugs and testing different scenarios and the memory function. Help: ChatGPT		
07/22	Restructured the GitHub repository and planning to transfer it to a new account (j-goth). Creating the documentation file. Begin working on the product documentation.	O, D	30
07/30	Minor changes to the files used to create game elements and group related files into separate folders. Restructuring the CMake setup due to problems encountered during compilation or linking. Help: ChatGPT	P, O	60
08/06	Catching up on the journal entries from the past few days/weeks. Preparing the project to transfer it to the new GitHub account. Resolving the linker issue by removing an incorrect “inline” from a function definition.	P, O	30
08/07	Continue writing the documentation for the “Product” chapter.	D	180
08/08	Continue writing the documentation for the “Product” chapter.	D	180
08/29	Continue writing the documentation and begin the introduction.	D	60

08/30	Completion of the text portion of the draft version of the documentation. Provisional submission of the unfinished version—with missing images and sources—as a precaution.	D	500
08/31	Completing the first draft and submit it to Mr. Leuthardt.	D	180
09/29	Beginning of the revision of the documentation based on the feedback.	D	30
09/30	Continue working on the revision.	D	30
10/03	Completed the revision of the documentation text. Adapted the program for compilation on Windows. Reduced the required dependencies by statically linking SDL3 and nlohmann-json → now, to run the compiled file, only Vulkan needs to be installed. Added SDL3 as a submodule of the Git project and adjusted the CMake configuration. Help: ChatGPT	D, P	300
10/04	Revised the remaining elements of the documentation (images, layout, etc.). Tried to create virtual machines running Windows 11 and macOS to test the program on all systems. My first choice was Oracle VirtualBox, but it wasn't suitable due to a lack of Vulkan support. Switched to qemu/kvm, but that didn't work either.	D, P	270
10/05	Attempt to adjust the CMake configuration so that the Vulkan shaders are automatically compiled only when they have been edited. The approaches I've tried so far all still have errors. Catching up on last week's journal entries. Help: ChatGPT	D, P	210

10/06	Adding the ability to edit and add connections in a JSON file so that the program doesn't have to be recompiled constantly. Help: ChatGPT	P	360
10/07	Adding a terminal interface that allows you to view and modify simulation parameters.	P	600
10/08	Überarbeiten der Dokumentation und einfügen aller Quellen.	D	420
10/09	Revising the documentation. The German version is complete. Writing a brief abstract for the BioValley College Award application. Translating the entire documentation into English using DeepL. Help: DeepL	D	480
10/10	Revising the English version of the paper. Finalize the abstract and translate it into English using DeepL as well. Help: DeepL	D	240
10/11	Finalizing and revising the journal. Translating to English using DeepL. Publish the entire project on GitHub. Compile and upload the program for Windows and Linux (GitHub Release). Final submission. Help: DeepL	D	300

This version of the journal was translated from German using DeepL